Normalization Exercises:

Multiple choice

1. What is a primary goal of database normalization?

A. Store all information in one table
B. Reduce unnecessary duplication and improve data consistency
C. Eliminate all foreign keys
D. Increase the number of repeated values

2. How does the normalization approach typically begin?

A. By identifying entities and relationships before examining data
B. By creating indexes for every column
C. With a collection of data items or an existing table
D. By removing every composite key

3. A university cannot record a new course until a student enrolls in it. Which anomaly does this illustrate?

A. Insertion anomaly
B. Update anomaly
C. Deletion anomaly
D. Transitive dependency

4. An instructor's office appears in several rows. Only some rows are changed when the instructor moves. Which anomaly results?

A. Insertion anomaly
B. Deletion anomaly
C. Update anomaly
D. Partial dependency

5. Which design violates First Normal Form?

A. A student has a separate row for each phone number
B. Each student has a unique Student ID
C. Each cell contains one value
D. One Email cell contains two emails

6. An enrollment table has the composite key (StudentID, CourseID). StudentName depends only on StudentID. What is this dependency called?

A. Transitive dependency
B. Partial dependency
C. Full dependency on the composite key
D. No dependency

7. Which dependency illustrates the problem addressed by Third Normal Form?

A. StudentID → StudentName
B. CourseID → CourseTitle
C. StudentID → DepartmentID → DepartmentName
D. (StudentID, CourseID) → Grade

8. Which statement about 3NF is correct?

A. It guarantees that no database can have any anomaly
B. It requires all data to be stored in one table
C. It removes the need for keys
D. It is sufficient for many practical databases, but some require higher normal forms

True or false

9. Deleting the last enrollment in a course can cause a deletion anomaly if it also removes the only stored information about that course.

10. A table must be in 1NF before it can be in 2NF.

11. A table is in 2NF when a non-key attribute depends on only part of a candidate key.

12. A table can be in 2NF and still contain a transitive dependency that violates 3NF.

13. Normalization eliminates every repeated value, including repeated foreign key values.

Short answer

14. Explain the difference between an insertion anomaly and a deletion anomaly.

15. Why can storing an instructor's office in every student enrollment row make the database harder to maintain?

16. Explain the difference between a partial dependency and a transitive dependency.

Application questions

17. Converting to 1NF

Consider this table:

StudentID	StudentName	PhoneNumbers
1001	John	555-1111, 555-2222
1002	Maria	555-3333

Explain why the table violates 1NF. Redesign it using a Students table and a StudentPhones table, and identify their primary keys.

18. Converting to 2NF

Consider this table:

StudentID	CourseID	StudentName	Grade
101	C01	Alice	A
101	C02	Alice	B
102	C01	Bob	B

Assume the only candidate key is **(StudentID, CourseID)**, each student has one name, and each enrollment has one grade.

Identify the partial dependency. Then split the table into tables that satisfy 2NF and identify their primary keys.

19. Converting to 3NF

Consider this table:

StudentID	StudentName	DepartmentID	DepartmentName
101	Alice	D01	Computing
102	Bob	D01	Computing
103	Carol	D02	Business

Assume StudentID is the only candidate key, each student belongs to one department, and each department has one name.

Identify the transitive dependency. Redesign the table to satisfy 3NF and identify the primary and foreign keys.

20. Identifying anomalies

A university uses one table:

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Enrollment(StudentID, CourseID, CourseTitle, InstructorID, InstructorOffice)
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Its primary key is **(StudentID, CourseID)**. Each course has one title and one instructor, and each instructor has one office.

Describe how this design could cause:

- An insertion anomaly
- An update anomaly
- A deletion anomaly

Propose a set of tables that would address these problems.